

Calculation Workbook - t6_r1_singlemodel

Sheet: final_metrics

Metric	Value	Unit	Criterion	Verdict
Volumetric energy density	1107.02	Wh/L	≥ 950	PASS
Discharge midpoint voltage	4.2078	V	≥ 4.1	PASS
Nominal capacity	5.4142	Ah	-	-
SEI thickness after 100 cycles	133.56	nm	≤ 500	PASS
4C charge T_max (45 degC ambient)	321.442	K	≤ 323.15	PASS
4C charge anode potential min	0.0454	V	≥ 0 (no plating)	PASS
4C charge duration to 4.7 V	64.7	s	-	-
4C charge fill	1.33	%	-	-
Cell mass	47.4	g	-	-
Cell thickness	199	um	-	-
DC resistance (calc-energy output)	5.39307e-07	ohm	-	-

Sheet: mechanism

Quantity	Value	Source
$\eta_e(\kappa)$ fit	$0.06868/\kappa + 0.00546$	diag_f3 trace (the F3 charge)
anode_pot = OCP_n - shift	shift = $0.615 * \eta_e$	diag_f3 decomposition
razor margin at the trip	$0.385 * \text{OCP}_n + 0.615 * \text{OCP}_c - 2.8856$	v4 plan algebra
V rising edge (t_ch 0-100 s)	$4.5515 + 0.00134 * t + \Delta \eta$	diag_f3 + the R10 fits
discharge fraction	$0.73 * Q_c$ (the cathode kinetic saturation)	diag_e5
clean V ceiling (the κ 2.0)	$4.696 - 0.092 + 0.05 = 4.654 \text{ V}$	diag_e5
measured η_e at the κ 0.19	0.3669 V	diag_f3
measured η_e at the κ 0.175 (the fit)	0.3979 V	the fit